SESSION 4: INTRODUCTION TO MS EXCEL-II

SESSION OVERVIEW:

By the end of this session, the students will be able to:

- Understand **conditional formatting** in detail.
- Understand how **cell referencing** works.
- Perform different **functions** in MS Excel.
- Understand different **data AI tools** in MS Excel.

KEY TOPICS AND EXAMPLES:

1. Conditional formatting: (Dataset)

➢ **Predefined Rules / Built-in Rules**

a. Uses:

i. Highlight data based on specified conditions
ii. Aiding in data analysis
iii. Visualization
iv. Error spotting
v. Creating visually appealing reports or dashboards.

b. Steps:

i. Select the data range.
ii. Go to the "Home" tab.
iii. Click "Conditional Formatting."
iv. Choose a rule from the dropdown menu.
v. Set the condition and formatting style.
vi. Click "OK" to apply the formatting.

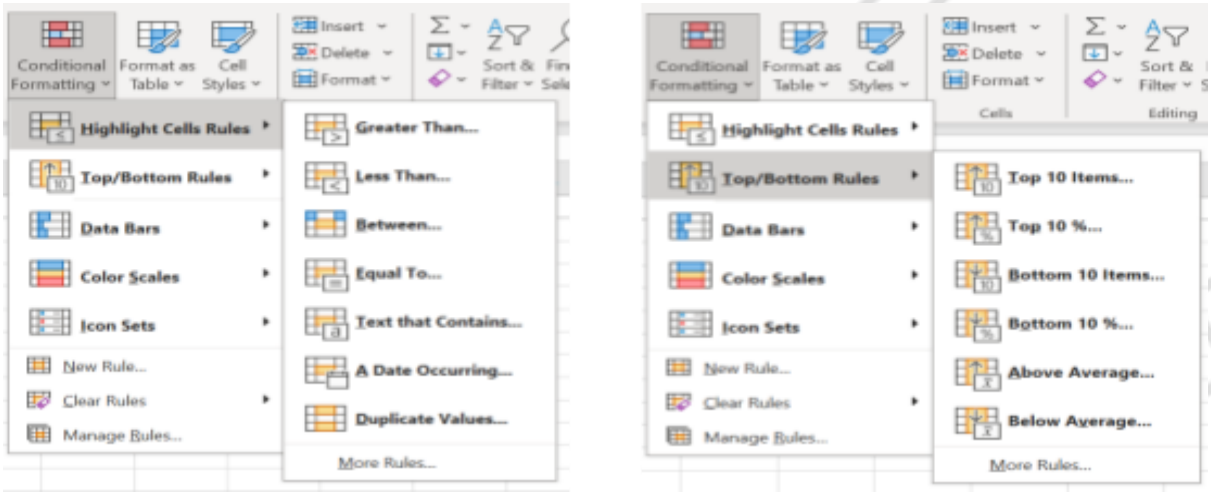


Figure 1: Representation of different conditional formatting-1

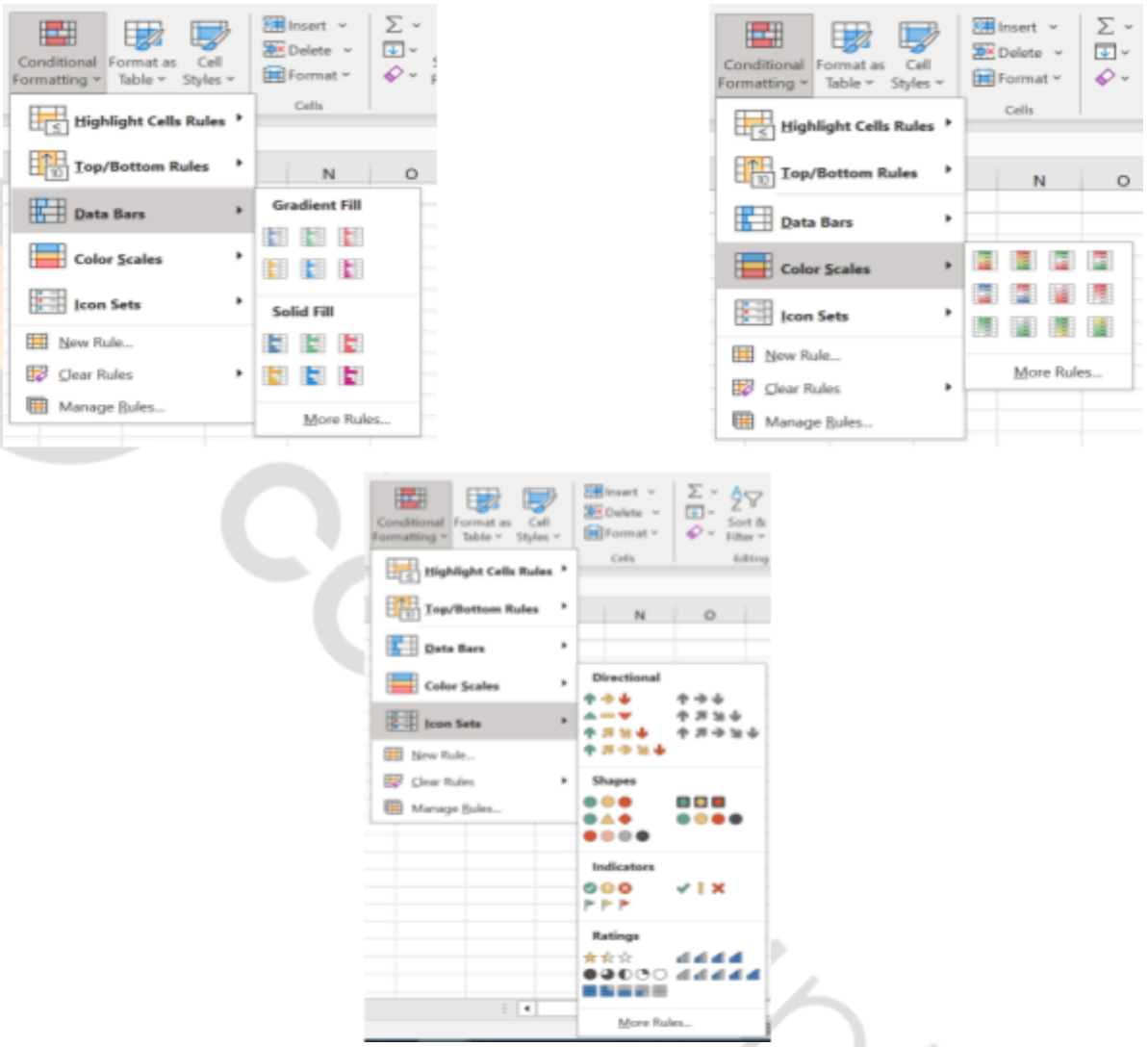
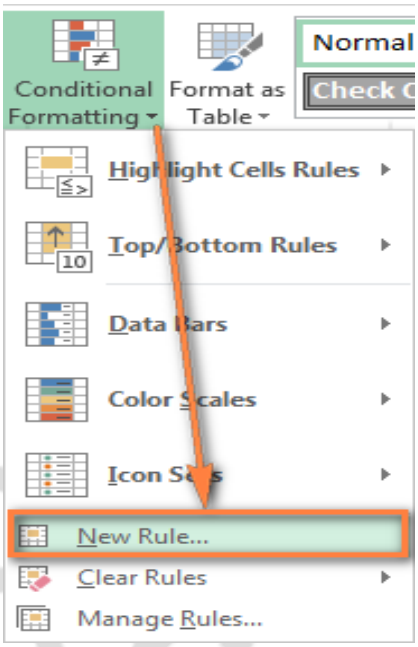


Figure 2: Representation of conditional formatting-2

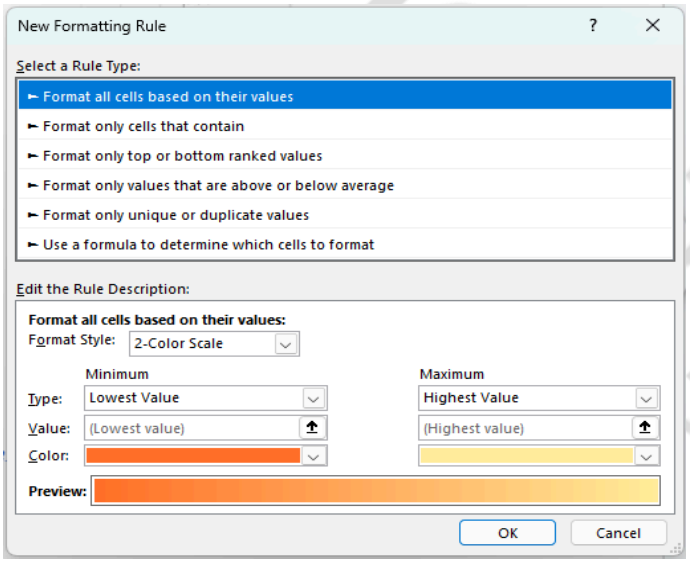
➢ New Rules (Custom Rules)

1. **Creating New Conditional Formatting Rules**
   - Go to Home → Conditional Formatting → New Rule.
   - Choose a rule type, such as:
     - Format cells based on values (e.g., highlight sales above a certain threshold).
     - Use a formula (e.g., highlight duplicate values using =COUNTIF(A:A,A1)>1).
   - Set formatting options like font color, fill color, or borders.
2. **Managing Existing Conditional Formatting Rules**
   - Go to Home → Conditional Formatting → Manage Rules.
   - Here, you can:
     - Edit Rules: Modify conditions or formatting styles.

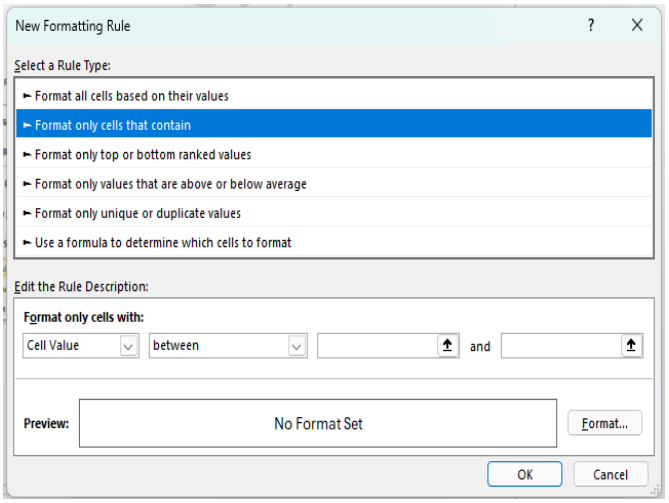
- Delete Rules: Remove outdated formatting.
- Change Rule Priority: Move rules up or down to control which one applies first.



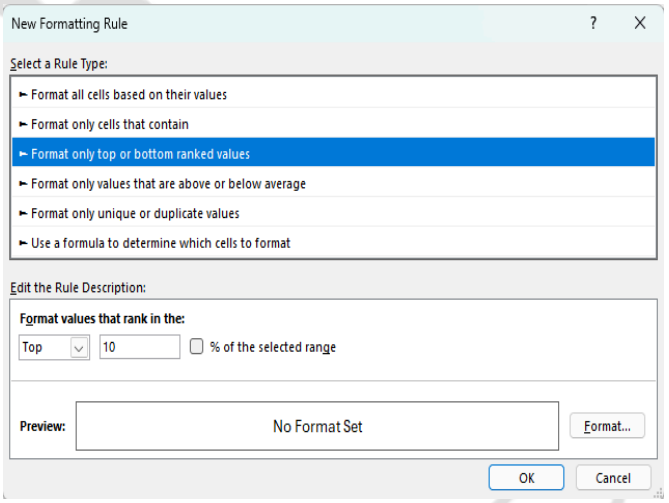
1. **Format all cells based on their values** – Applies color scales, data bars, or icon sets dynamically based on cell values.



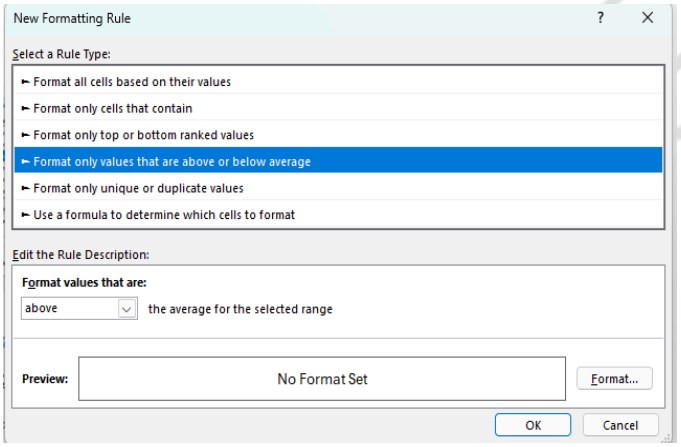
2. **Format only cells that contain** – Highlights specific cells meeting conditions like greater than, equal to, or text-based criteria.



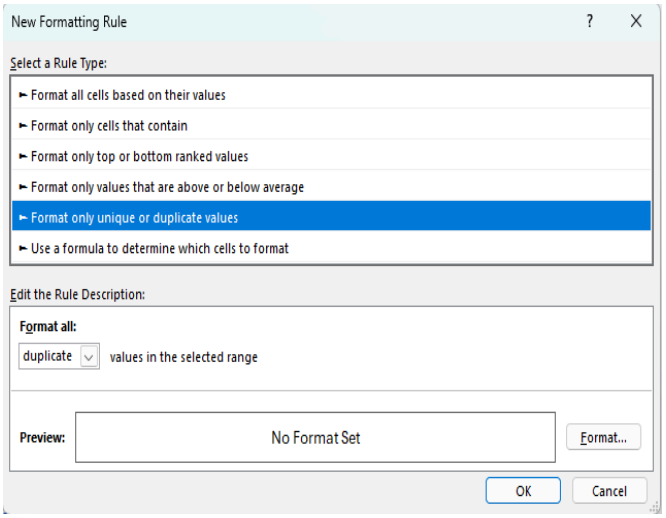
3. **Format only top or bottom ranked values** – Emphasizes the highest or lowest values in a range, useful for rankings.



4. **Format only values that are above or below average** – Shades cells based on deviations from the dataset's average.



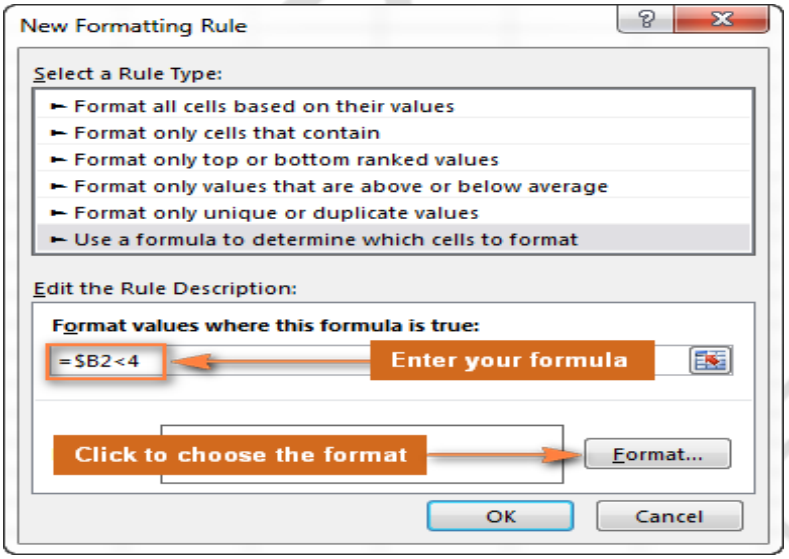
5. **Format only unique or duplicate values –** Distinguishes repeated entries or highlights unique values automatically.



6. **Use a formula to determine which cells to format** – Offers full customization, allowing complex conditions using Excel formulas

How It Works in Conditional Formatting:

Example1. If the condition **$B2 < 4** is true for any row, Excel formats that row as specified.



Example2. The formula =AND($C2>0, $D2="Worldwide") in **Conditional Formatting** checks if:

- Column **C** (Sold) has a value greater than **0**
- Column **D** (Delivery) equals **"Worldwide"**

If both conditions are true, the row is highlighted, making it easier to track globally sold products. The applied range **($A$2:$E$8)** ensures only relevant data is formatted.

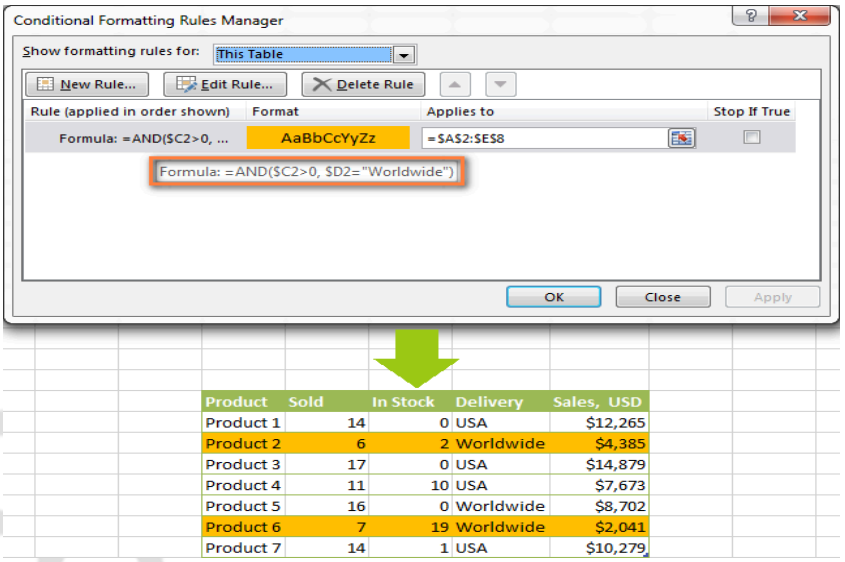


***Figure 3**: Representation of conditional formatting-3 (New Formatting Rule)*

2. Understanding cell referencing in MS Excel:

a. Relative referencing: (Dataset)

- Relative referencing in Excel refers to a method where cell references in formulas are adjusted based on their relative position to the cell containing the formula when copied to other cells.
- Relative referencing is the default behavior in Excel, and it's useful for creating formulas that can be easily copied and applied to different parts of a worksheet without having to manually adjust cell references.

Example: In the example mentioned below, we can see that in the formula the cells are automatically being adjusted.

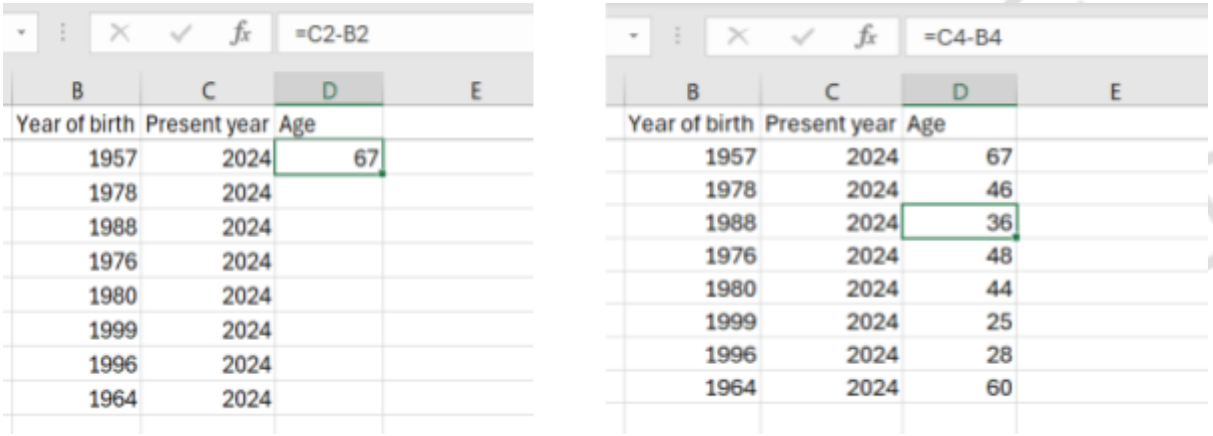


***Figure 25:** Representation relative referencing*

b. **Absolute referencing: (Dataset)**

- Absolute referencing in Excel refers to a method where cell references in formulas are fixed and do not change when the formula is copied to other cells. This means that the reference to a specific cell remains constant regardless of where the formula is copied within the worksheet.
- In absolute referencing, cell references are denoted by adding a dollar sign ($) before the column letter, row number, or both.
- Absolute referencing is useful when you want to refer to a specific cell or range of cells that should not change when copying formulas to other parts of the worksheet. This ensures that the formula always refers to the same cells, regardless of its position within the worksheet.

c. **Mixed Referencing**

- Mixed referencing is a type of cell reference where either the row or the column is fixed, but not both. This is the middle ground between relative and absolute referencing, allowing one part of the coordinate to stay constant while the other part remains free to change when the formula is copied.

How it Works:

- In mixed referencing, the dollar sign ($) is placed only before the part of the reference you want to "lock":
- Fixed Column ($C2): The dollar sign is before the column letter. If you copy the formula to the right, the reference stays on Column C. If you copy it down, the row number changes (2, 3, 4...).
- Fixed Row (C$2): The dollar sign is before the row number. If you copy the formula down, the reference stays on Row 2. If you copy it to the right, the column letter changes (C, D, E...).

d. **Using the F4 Key in Excel**

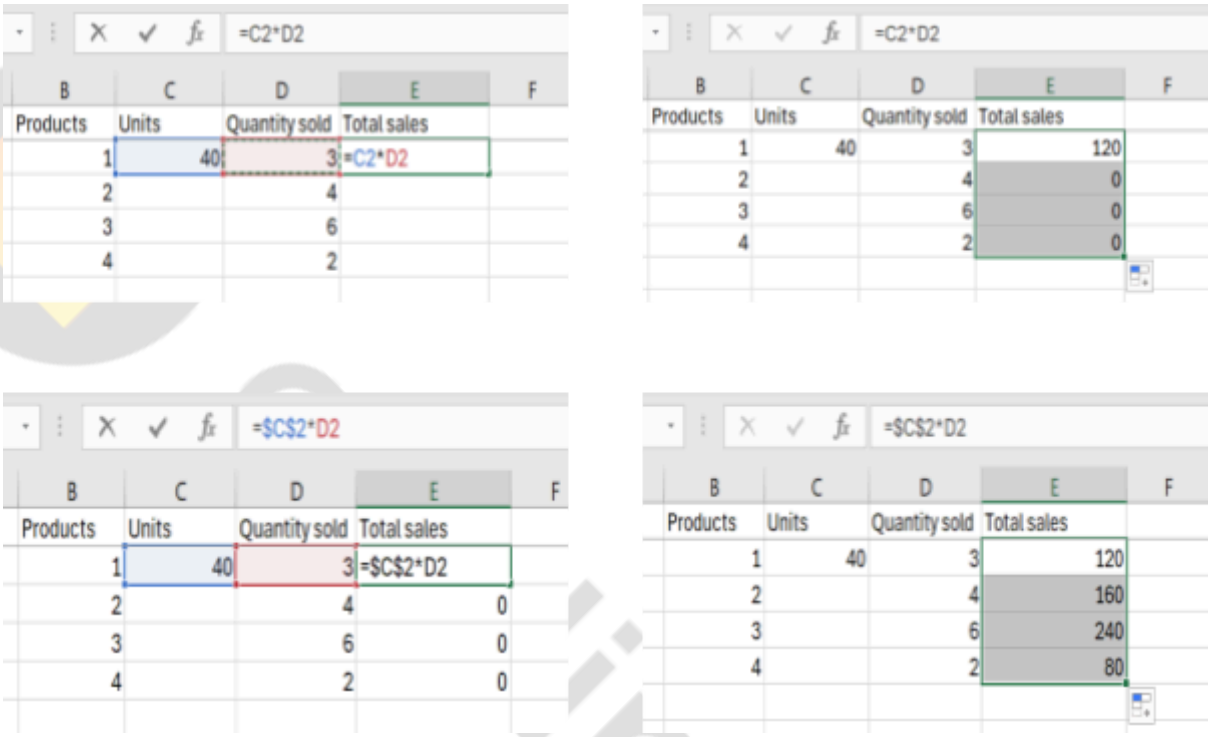
Here's how the F4 key cycles through the reference types:

- **A1 → Relative reference:** Both row and column will change when copied to other cells.
- **$A$1 → Absolute reference:** Both row and column are locked; they do not change when copied.
- **A$1 → Mixed reference:** The row is locked, but the column changes.
- **$A1 → Mixed reference:** The column is locked, but the row changes.

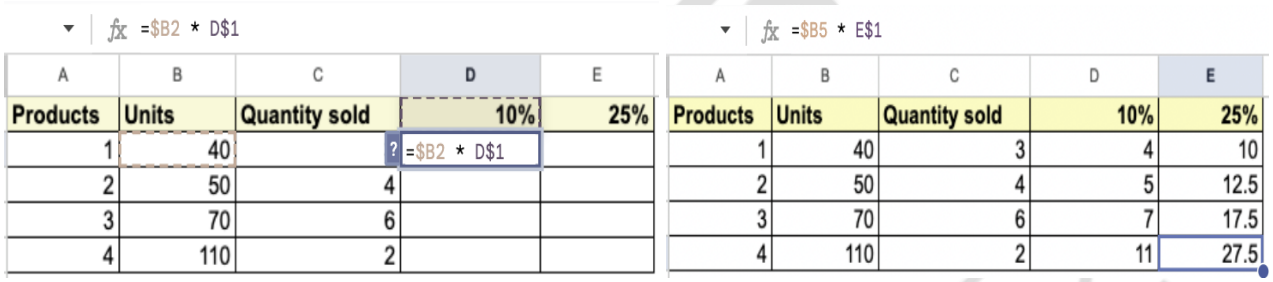
This makes F4 extremely important when creating formulas that rely on a constant value (e.g., a fixed tax rate or interest rate in one cell) while still applying calculations to multiple rows or columns of data.

Example: The below-mentioned is a hypothetical example of total sales.

1. The first diagram shows the utilization of the formula in the data.
2. The second diagram shows the formula not working in relative referencing.
3. The third diagram represents the use of $ for absolute referencing.
4. The fourth diagram represents the results after using $.



(Absolute Referencing.)



(Mixed Referencing)

Example : (Absolute Reference)

Question:

Cell A2 contains the price of an item. Cell B1 contains a fixed tax rate (10%). Write a formula to calculate the price including tax in B2, and make sure the tax cell stays constant when copied down.

Solution:

Formula:

```
=A2*$B$1
```

How it works:

- $B$1 is an absolute reference (both row and column locked).
- Copying down will change A2 → A3, A4…, but $B$1 will always point to the tax rate.

Example : (Mixed Reference - Row Locked)
Question:
You want to multiply the numbers in row 1 (headers) with employee sales in column A. Write the formula in B2 so that the row (header) stays fixed when copied across.

Solution:
Formula:

```
=B$1*$A2
```

How it works:

- $1 keeps the row fixed while copying across.
- $A2 keeps the column fixed while copying down.
- This uses mixed references.

Example : (Mixed Reference - Column Locked)
Question:
You are preparing a sales report. The commission rate is written in row 1 of column D. Write a formula in B2 so that the formula always refers to column D while copying across, but allows the row number to adjust when needed.

Solution:
Formula:

```
=B2*D$1
```

How it works:

- D$1 → column is fixed (D), but row can change.
- This is a mixed reference.

Figure 26: Represents Absolute referencing

3. A. Importance of Functions in MS Excel:

- **Automating Calculations**: Functions allow users to automate complex calculations, saving time and reducing the risk of errors.

- **Data Analysis**: Excel functions facilitate data analysis by performing statistical, financial, and mathematical calculations, enabling users to derive insights from their data.
- **Decision Making**: Logical functions help users make decisions based on specified criteria, allowing for conditional formatting, data validation, and dynamic reporting.
- **Data Manipulation**: Functions enable users to manipulate and transform data, such as extracting text, splitting strings, or combining values from different cells.
- **Customization**: Excel functions provide users with the flexibility to customize formulas and functions based on their specific requirements, allowing for tailored solutions to unique problems.
- **Efficiency:** By leveraging built-in functions, users can accomplish tasks more efficiently, enhancing productivity and streamlining workflow processes.

4. B. Disadvantages of Functions in MS Excel:

- **Complexity**: As Excel offers a vast array of functions, it can be overwhelming for users to understand and remember all available functions and their syntax.
- **Learning Curve:** Learning how to use functions effectively requires time and effort, especially for complex functions or formulas.
- **Errors**: Misusing functions or inputting incorrect parameters can lead to errors in calculations, potentially resulting in inaccurate results or decisions.
- **Performance Issues**: Complex formulas or functions can slow down spreadsheet performance, especially in large datasets or files with many calculations.
- **Limited Functionality**: Excel functions have limitations, and some tasks may require more advanced programming or scripting languages to accomplish.
- **Version Compatibility**: Functions may behave differently across different versions of Excel, leading to compatibility issues when sharing files between users with different versions.

5. Perform different functions in MS Excel: (dataset)

- **Mathematical Calculations**: Functions like SUM, AVERAGE, MAX, and MIN are used for basic arithmetic operations and statistical calculations.
- **Statistical Analysis:** Functions like COUNT, COUNTA, COUNTIF, and SUMIF are used for counting and summing data based on specified conditions, as well as other statistical calculations.
- **Logical Operations:** Functions like IF, AND, OR, and NOT are used to perform logical operations and make decisions based on specified criteria.
- **Text Manipulation:** Functions like CONCATENATE, LEFT, RIGHT, and MID are used to manipulate and concatenate text strings. These operations are majorly used in data cleaning before the data analysis.
- **Date and Time Operations**: Functions like TODAY, NOW, DATE, and TIME are used to work with date and time values and perform calculations based on them.

a. **Mathematical functions: (Dataset)**

Steps:

- Click on the cell where you want the result of the function to appear.
- Start typing the name of the function you want to use (e.g., SUM, AVERAGE, MIN, MAX).
- After typing the function name, open parentheses "(" to begin entering arguments.
- Enter the range of cells or values you want to include in the calculation. Separate multiple arguments with commas.
- Close parentheses ")" to indicate the end of the function's arguments.
- Press Enter to complete the function. Excel will calculate the result based on the specified function and arguments.

i. **Sum**: Adds up numbers in a range.

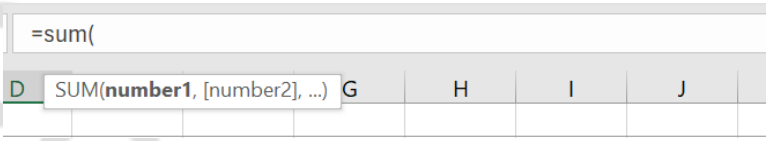


Figure 3: *Representation of SUM function in Excel*

ii. **Average**: Calculates the average of numbers in a range.

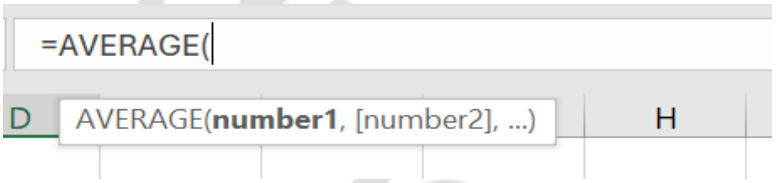


Figure 4: *Representation of the AVERAGE function in Excel*

iii. **MAX:** Finds the maximum number in a range.

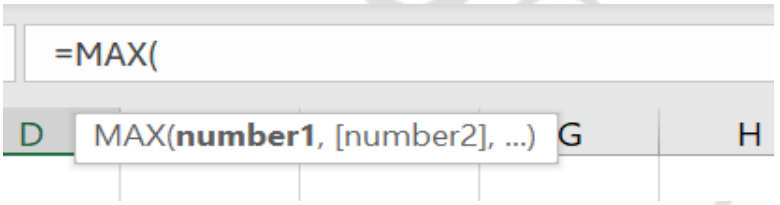


Figure 5: *Representation of MAX function in Excel*

iv. **MIN:** Finds the minimum number in a range.

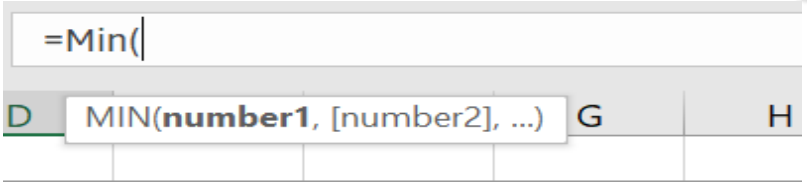


Figure 6: *Representation of the MIN function in Excel*

In-Class Practice Activity:

Dataset - employee_sales.csv

Problem: The Finance Department's "Operational Drag" Audit

Management at Elite Dynamics is investigating "Operational Drag"—valuable funds tied up in transactions that are not yet finalized. You have been tasked with auditing the Finance Department to identify high-volume specialists who have Pending orders.
Your goal is to pinpoint specific financial risks. However, management is only interested in employees whose individual work volume (Quantity Sold) is higher than the average volume of their department peers.
To complete the audit and find the "Peak Risk" employee:

- Identify Performance Outliers: Locate transactions within the Finance department that remain in a "Pending" state and where the specialist's individual Quantity Sold exceeds the departmental average.
- Assess Financial Impact: Apply a standard 15% Processing Surcharge to the Revenue of these specific transactions. To maintain audit integrity and allow for global updates, you must house the 0.15 rate in cell M1 and ensure your surcharge formula uses an Absolute Reference to that cell.
- Identify Critical Flags: Apply a visual indicator to the surcharge results to call out any amounts that surpass the $1,000 threshold.

Based on your multi-step audit, who is the employee with the single highest highlighted surcharge, and what is the total number of Finance employees flagged by the $1,000 visual rule?

OPTIONS:

A) Jessica Smith; $1,337.70; 3 Employees
B) Lauren Daniels; $1,222.20; 2 Employees
C) Margaret Hawkins; $894.60; 5 Employees
D) Jessica Smith; $1,337.70; 3 Employee

Expected Answer: Option A (Jessica Smith; $1,337.70; 3 Employees)